

Real-Time Post-Market Signal Integration for Enhanced Surveillance

April 1, 2026

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Vendor Day Summary

The FDA hosted a vendor day to support ongoing market research for its Real-Time Post-Market Signal Integration for Enhanced Surveillance initiative. The session brought together industry stakeholders to clarify the intent of the RFI, address key questions, and gather insight into current industry capabilities.

The core objective of this effort is to **shift FDA's post-market surveillance approach from reactive to proactive and predictive**, enabling earlier identification of safety signals across all FDA-regulated products.

As emphasized during the session, current surveillance approaches often result in significant delays (months to over a year) between event occurrence and detection. This initiative aims to dramatically reduce that timeline, with a focus on identifying signals within a 24-hour window or sooner where feasible.

Rather than relying solely on internal FDA data systems, the Agency is exploring how external data sources and industry-generated signals can enhance earlier detection of emerging risks.

Key themes from the discussion included:

- Moving toward real-time or near real-time signal detection
- Leveraging diverse, external data sources across industry
- Focusing on early awareness of emerging risks, even before full validation
- Enabling cross-domain signal correlation (e.g., combining EHR, social, claims, etc.)
- Prioritizing public health impact and faster time to action

The FDA reinforced that this effort is **signal-driven**, not platform-driven—vendors are not being asked to provide full systems, dashboards, or hosted environments, but rather **actionable signals derived from their data and analytics capabilities**.

Q&A Summary

The following summarizes key themes and clarifications discussed during the vendor day Q&A session. These highlights are intended to provide a high-level understanding of FDA's current thinking and areas of interest.

For the complete set of consolidated questions and additional context, please refer to the vendor day presentation materials.

1. Definition of Real-Time / Near Real-Time

- FDA considers "near real-time" as within ~24 hours of signal detection.
- Faster (e.g., streaming or continuous detection) is encouraged where possible.
- Goal: identify signals early enough to intervene before escalation.

2. Current Detection Timelines & Success Metrics

- Current timelines are not sufficient and can take months.
- Success will be measured by:
 - Earlier detection
 - Faster time to action
 - Reduction in harm (injuries/deaths)
- Emphasis on public health outcomes, not just technical performance.

3. Data Sources & Signal Expectations

- FDA is not seeking reuse of its own datasets (e.g., FAERS, MAUDE).
- Vendors are expected to:
 - Leverage their own data sources
 - Detect and send signals to FDA
- Examples of desired signals:
 - Safety events (e.g., adverse reactions)
 - Product performance issues
 - Supply chain or manufacturing concerns
- FDA is interested in both:
 - Confirmed signals
 - Emerging signals (early trends, rising patterns)

4. Emerging Signals & Signal Evolution

- FDA is highly interested in early-stage signals, even below traditional thresholds.
- Signals may:
 - Start as small clusters (e.g., 3–5 cases)
 - Become meaningful through trend acceleration
- Cross-source validation (e.g., social + clinical data) strengthens signals.

5. Level of Detail Required

- FDA prefers:
 - Minimal necessary data (“least but no less”)
 - Aggregated or summarized signal data
- Avoid PII/PHI unless explicitly required
- Focus is on:
 - Signal existence
 - Event type
 - Counts/trends
 - Timing

6. Data Ownership & Architecture

- FDA does NOT want to host vendor data or systems.

- Model is:
 - External vendor analysis
 - Signals sent to FDA via API
- Anticipated approach: One-way communication (vendor → FDA)
- Infrastructure choices (cloud, on-prem, etc.) are up to vendors.

7. Signal Delivery Model

- Preferred: API-based, event-driven signal delivery
- Not preferred:
 - Batch uploads
 - Full platform integrations
- FDA is not seeking dashboards, visualization tools, or hosted environments.

8. Validation & False Positives

- Vendors should:
 - Describe how signals are generated and validated
 - Provide insight into confidence levels and false positive reduction
- FDA will perform final prioritization and ranking of signals internally

9. Cross-Domain Signal Correlation

- Strong interest in:
 - Signals derived from multiple data sources
 - Corroborated signals across domains
- Vendors with multi-source data or correlation capabilities are encouraged to highlight this.

10. Signal Processing & FDA Role

- Vendors:
 - Detect and submit signals
- FDA:
 - Aggregates, validates, prioritizes, and acts on signals
- FDA will not rely on vendor platforms for analysis or triage

11. Acquisition Strategy & Vendor Model

- No final acquisition strategy yet.
- FDA anticipates multiple vendors, not a single provider
- Vendors should propose pricing models (e.g., per signal, subscription, volume-based)

12. Signal Detection Methods

- Vendors are responsible for their own algorithms and methodologies

- FDA is open to:
 - AI/ML
 - Traditional analytics
 - Novel approaches
- No prescribed methodology.

13. Follow-On Analysis

- FDA is not currently planning iterative vendor-driven investigations after signal submission.
- Open to ideas if vendors propose viable approaches.

14. Scope of Signals

- Applies to ALL FDA-regulated products, including:
 - Drugs
 - Devices
 - Biologics
 - Food
- Primary focus for this RFI:
 - Primary signals (not derived/secondary signals)

Key Takeaways for Industry

- FDA is building a **signal-driven ecosystem**, not a centralized data platform.
- The Agency is looking for:
 - Speed
 - Breadth of data
 - Early detection
 - Actionable insights
- Vendors should focus on:
 - What signals they can detect
 - How quickly they can detect them
 - How confidently they can validate them
- Innovation is encouraged—especially in:
 - Cross-domain analytics
 - Emerging signal detection
 - Real-time data processing

Next Steps

- RFI responses are due: **April 7, 2026, at 10:00 AM**
- Submissions must follow instructions outlined in the SAM.gov posting.
- FDA will continue market research and determine next steps based on responses.